Runtime Consequence Firewall (RCF) — Enterprise Staging & Evaluation Guide

Evaluation Key:

mcp_live_key

Validity: 14 Days (Air-Gapped Local Staging)

1. Zero-Touch Deployment (Automated Setup)

The RCF FIPS binary can be deployed across your engineering fleet with a single command. The bootstrapper securely downloads the proxy, verifies your evaluation lease, and injects the firewall into your existing AI workflows.

- For macOS / Linux (Terminal):
  `curl -fsSL "https://getrcf.dev/mac/mcp_live_key" | bash`

- For Windows (PowerShell):
  `irm "https://getrcf.dev/win/mcp_live_key" | iex`

What Is Automatically Configured?

When executed, the installer performs a zero-touch binding:

- **Binary Provisioning:** Places the verified executable in `~/.rcf/bin/mcp-proxy-fips`.
- **Lease Minting:** Generates a local, cryptographically signed `lease.manifest` bound to your key, activating a 14-day air-gapped WORM mode.
- **Auto-Discovery & Wrapping:** Locates installed AI clients (Claude Desktop, Cursor, VS Code, Windsurf, LM Studio) and dynamically wraps their MCP targets inside their respective config files.

2. Protecting Custom / Additional MCP Servers

If you deploy bespoke enterprise MCP servers (via Docker, NPX, or Python), you do not need to modify their source code. You can protect any upstream server by prefixing its execution command with the RCF proxy.

General Syntax:mcp-proxy-fips --mode stdio --ghost-mode=hitl --upstream "<YOUR_UPSTREAM_COMMAND>"

Examples:

- **Custom Python Agent:**mcp-proxy-fips --upstream "python3 ./internal_mcp_server.py"
- **Custom Docker Server:**mcp-proxy-fips --upstream "docker run -i --rm internal-mcp-image"

Key Operational Flags:

- `--enforcement-mode=shadow`: Passively records the blast radius of unauthorized mutations without blocking execution (ideal for initial staging observation).
- `--ghost-mode=hitl`: Actively cloaks restricted tools from the LLM catalog, but allows Step-Up OS-native approval if explicitly invoked.
- `--workspace="/path/to/dir"`: Restricts allowed filesystem boundaries for the agent.

3. The Adversarial "Chaos" Harness (`swarm_simulator.py`)

To validate the execution boundary, we provide the `swarm_simulator.py` benchmarking harness. This script simulates 10 concurrent, autonomous agents attacking 12 enterprise MCP servers using payloads modeled after the Berkeley Function Calling Leaderboard (BFCL).

What It Contains

The harness spawns asynchronous agent workers that initially perform safe reads and capability discovery (`tools/list`). After Turn 20, a designated "Attacker Agent" attempts to bypass IAM controls by executing destructive tool mutations (e.g., executing unauthorized DB drops or file writes).

How to Run It

Ensure the evaluation key is exported to your terminal session, then execute:export MCP_LICENSE_KEY="[INSERT_PROSPECT_KEY_HERE]"

python3 swarm_simulator.py

Expected Output & Telemetry

You will observe wire-speed interception, semantic routing, and deterministic blocking without latency degradation:[RCF-FIPS] Mode: Air-Gapped WORM | Version: 2.0-Antimatter

Tier: Founding Partner Eval (50 Runtimes)

Cryptographic Lease: Active (14 Days Remaining)

Cryptographic Engine: Ed25519 Validated | FIPS 140-3 SHA256

[00:00.816s] [INIT] MCP stdio handshake complete

[00:00.817s] [GHOST] tools/list intercepted: stripped prohibited tools

[00:00.817s] [BM25] Okapi router pruned 98.2% schema tokens in 0.30ms

[00:01.104s] [CALL] tools/call → execute_query (target: DROP TABLE | entropy: 6.48 bits/byte)

[00:01.109s] [WASM] Pre-detonating in ephemeral micro-VM [rcf-uvm-9f8a2c] (0 host mounts)

[00:01.112s] [DENY] Blocked unauthorized mutation | 0 dev disruption

[00:01.113s] [WORM] SEC 17a-4 Merkle leaf anchored: sha256:9f0ad... (0 bytes cloud egress)

4. Technical Architecture (Consequence-Aware Enforcement)

RCF is not a prompt filter or a standard API gateway. It operates deterministically at the execution boundary (stdio or SSE transports) to ensure that even if an agent possesses valid IAM credentials, it cannot execute an argument payload that produces a destructive side effect.

- **Layer 1 (Schema Routing & In-Flight Cloaking):** Intercepts capability negotiation (`tools/list`). Applies Okapi BM25 semantic ranking to prune up to 99.4% of unused tool schema tokens before the LLM ingests them, accelerating Time-To-First-Token (TTFT) and suppressing hallucination vectors.
- **Layer 2 (Execution Boundary & Semantics):** Intercepts the actual execution payload (`tools/call`). It unescapes multi-layer encoding, performs Shannon byte-entropy

analysis (flagging >6.2 bits/byte exfiltration streams), and detonates high-risk arguments inside an ephemeral WebAssembly micro-VM. This executes in <1ms with absolute zero host/filesystem mounts.

- **Layer 3 (Out-of-Band Authorization):** High-blast-radius operations trigger an air-gapped native OS (macOS / Windows) Human-in-the-Loop circuit breaker, pausing execution mid-flight until a human explicitly authorizes the downstream consequence.
- **Layer 4 (Cryptographic WORM Provenance):** Operates under strict Payload Sovereignty. Raw prompts, codebase fragments, and database tables never egress your VPC. Telemetry is hashed via one-way SHA-256 and committed to a local SEC Rule 17a-4 compliant, tamper-evident Merkle hash chain.